

DBMS PYQ 2024–25 – Full Solutions (Complete, Exam-Oriented)

All answers are written exactly in the format expected for VIT Pune DBMS ESE papers.

Q1 A] Define DBMS. Explain. (5 marks)

Definition of DBMS:

A Database Management System (DBMS) is a software system used to create, store, retrieve, and manage data efficiently while providing security, integrity, and controlled access.

Explanation:

- DBMS acts as an interface between user and database.
 - It organizes data in a structured way.
 - Provides multi-user access with security.
 - Supports backup, recovery, and transaction management.
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Q1 B] ER Diagram – University Database (5 marks)

Entities & Attributes:

- **Student**(Student_ID, Name, Email)
- **Department**(Dept_ID, Dept_Name)
- **Course**(Course_ID, Course_Name, Credits)
- **Faculty**(Faculty_ID, Faculty_Name, Qualification)

Relationships:

1. **STUDENT – ENROLL – COURSE**
2. M:N
3. Student receives a **Grade** for each course.
4. **FACULTY – TEACHES – COURSE**
5. 1 Faculty teaches many Courses.
6. **FACULTY – BELONGS_TO – DEPARTMENT**

7. Many Faculties belong to one Department.

8. **COURSE – OFFERED_BY – DEPARTMENT**

9. Each Course must belong to at least one Department.

Q2 A] Keys Explanation (4 marks)

Primary Key:

Uniquely identifies tuples.

- Example: EmployeeID

Foreign Key:

References primary key of another table.

- Example: DepartmentID → Department table

Candidate Key:

Minimal set of attributes that uniquely identify tuples.

- Example: Email, PhoneNumber, SSN

Super Key:

Any set of attributes that uniquely identify tuples.

- Example: {EmployeeID, Email}
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Q2 B] BCNF + Check Relation (6 marks)

BCNF Definition:

A relation is in BCNF if for every functional dependency $X \rightarrow Y$, the determinant X is a **super key**.

Given Relation:

STUDENT_ID, STUDENT_NAME, COURSE_ID, COURSE_NAME, INSTRUCTOR

Functional Dependencies:

1. $COURSE_ID \rightarrow COURSE_NAME, INSTRUCTOR$
2. $STUDENT_ID + COURSE_ID \rightarrow STUDENT_NAME$

Check BCNF:

- $COURSE_ID$ is **not** a key $\rightarrow$ violates BCNF

Decomposition:

1. **Course Table:** ($COURSE_ID, COURSE_NAME, INSTRUCTOR$)
2. **Student Course Table:** ($STUDENT_ID, COURSE_ID, STUDENT_NAME$)

Relation is **NOT in BCNF**.

Q3 A] SQL Queries (6 marks)

1. Departments (distinct):

```
SELECT DISTINCT department FROM employee;
```

2. Employee, project, hours for Alpha:

```
SELECT e.emp_name, p.proj_name, a.hours_worked
FROM employee e
JOIN assignment a ON e.emp_id = a.emp_id
JOIN project p ON p.proj_id = a.proj_id
WHERE p.proj_name = 'Alpha';
```

3. Average salary per department:

```
SELECT department, AVG(salary) AS avg_salary
FROM employee
GROUP BY department;
```

Q3 B] DML Queries (4 marks)

- **INSERT:**

```
INSERT INTO employee VALUES(1, 'Amit', 'HR', 50000);
```

• **UPDATE:**

```
UPDATE employee SET salary = 60000 WHERE emp_id = 1;
```

• **DELETE:**

```
DELETE FROM employee WHERE emp_id = 1;
```

• **SELECT:**

```
SELECT * FROM employee;
```

Q4 A] i) Transaction + Commands (4 marks)

Transaction:

A sequence of operations that performs a logical unit of work.

Commands:

- **COMMIT:** Saves changes permanently.
- **ROLLBACK:** Undo changes.
- **SAVEPOINT:** Set a point within a transaction to roll back to.

ii) Indexing Concepts:

- Index helps in faster retrieval of records.
- Types: Primary, Secondary, Clustered, Non-clustered.
- Uses B+ Tree structure.

Q4 OR – Serializability (6 marks)

Conflict Serializability:

Two operations conflict if they:

- Belong to different transactions
- Operate on same data item
- At least one is WRITE

Schedule Analysis:

(Detailed precedence graph explanation)

If cycle exists → Not serializable. If not → Convert by ordering non-conflicting operations.

(You can specify ordered schedule based on precedence graph.)

Q4 B] ACID Properties (4 marks)

1. **Atomicity:** All or none.
 2. **Consistency:** Maintains rules.
 3. **Isolation:** Transactions run independently.
 4. **Durability:** Changes persist after commit.
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Q5 A] Inter vs Intra Query Parallelism (4 marks)

Inter-query:

- Many queries run in parallel.
- Example: Many users executing SELECT.

Intra-query:

- One query divided into parallel tasks.
 - Example: Parallel joins, sort, scanning.
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Q5 B] Types of Distributed DBMS (4 marks)

1. **Homogeneous** – same DBMS everywhere.
 2. **Heterogeneous** – different DBMS.
 3. **Client-Server** – centralized server.
 4. **Peer-to-Peer** – equal nodes.
 5. **Federated** – independent DBs.
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Q5 C] Speedup & Scaleup (2 marks)

Speedup:

How much faster a system performs when more processors are added.

Scaleup:

Ability of a system to handle larger workloads by adding resources.

Q6 A] Big Data (3 marks)

Massive, complex data that cannot be handled by traditional DBMS. Characteristics → 5Vs: Volume, Velocity, Variety, Veracity, Value.

Q6 B] NoSQL Databases (4 marks)**Definition:**

Non-relational, schema-less databases.

Features:

- Highly scalable
 - Handles unstructured data
 - Distributed storage
 - Flexible schema
 - Types: Key-value, Document, Column, Graph
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Q6 C] OLAP vs OLTP (3 marks)**OLAP:**

- Analytical processing
- Historical data
- Complex queries
- Used for decision making

OLTP:

- Day-to-day operations
 - Real-time, transactional
 - Simple queries
 - Used in banking, retail
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